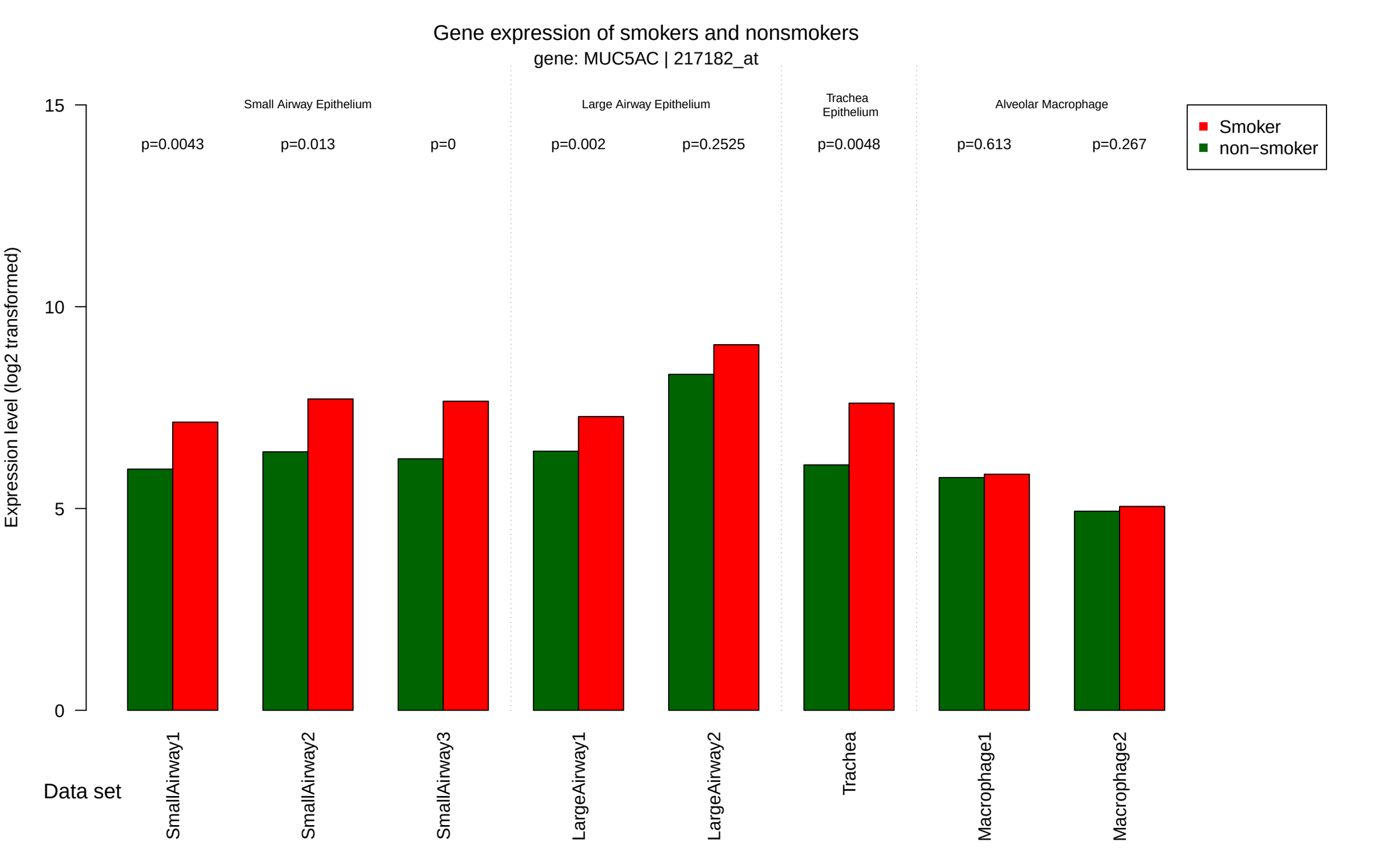
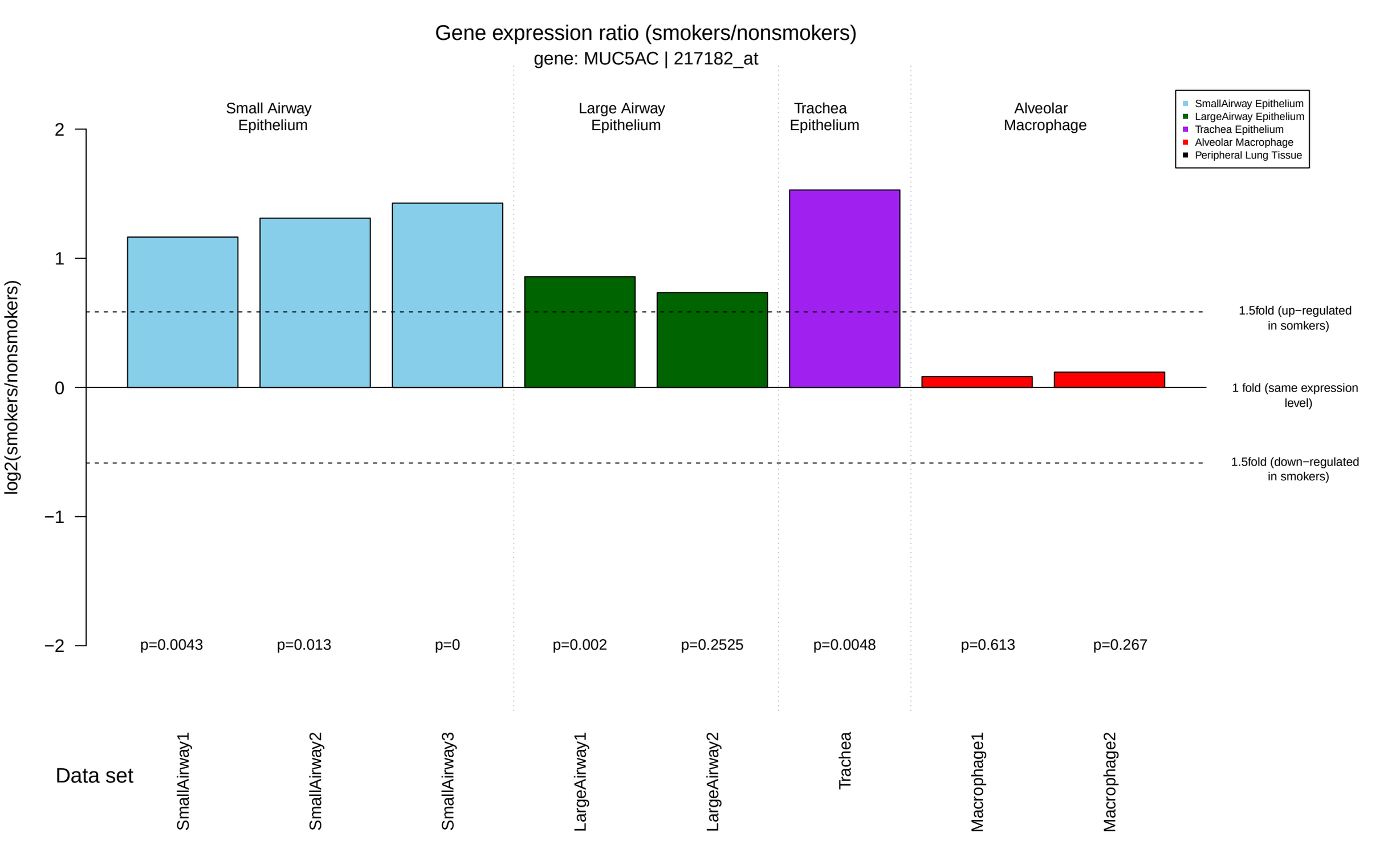
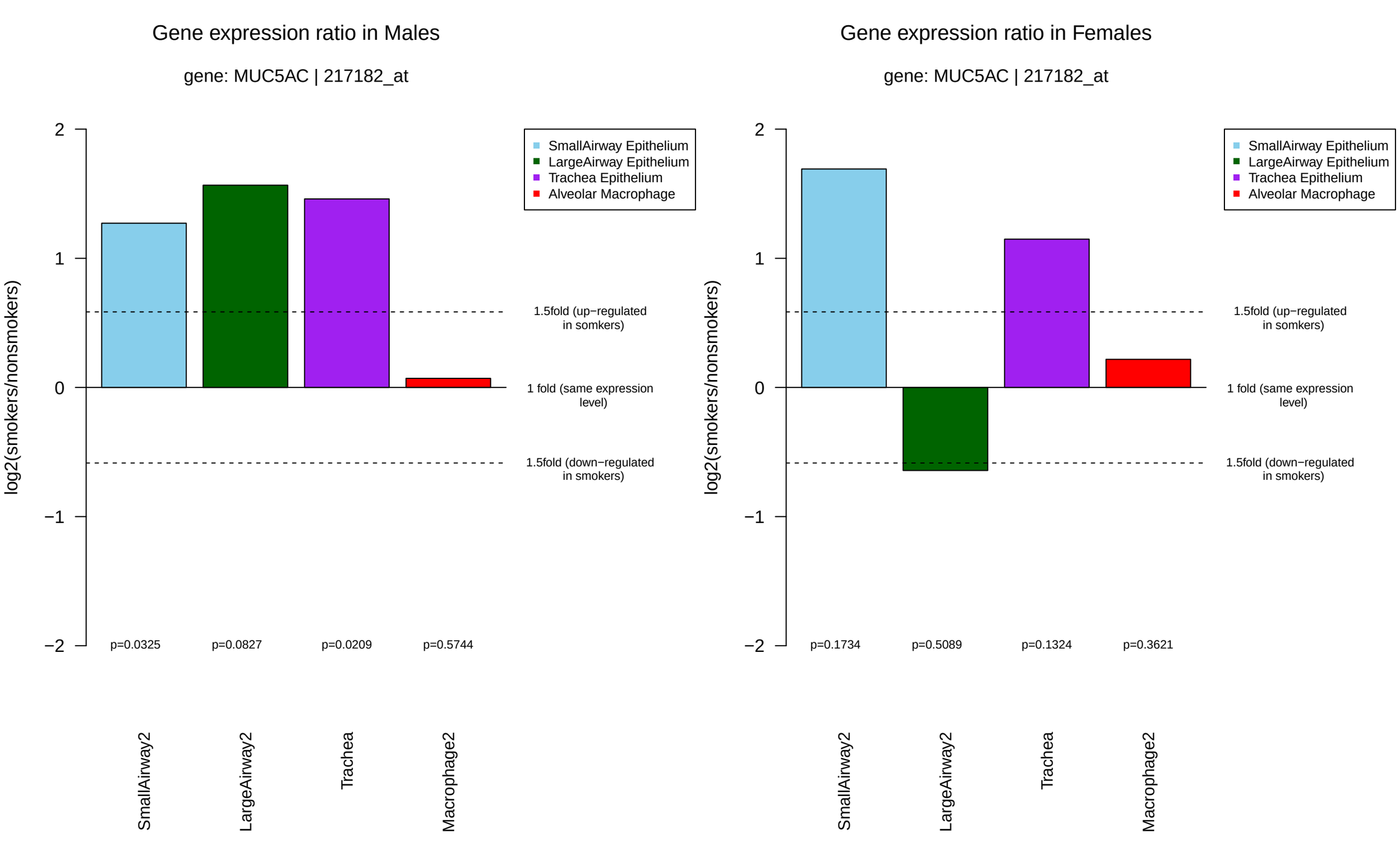
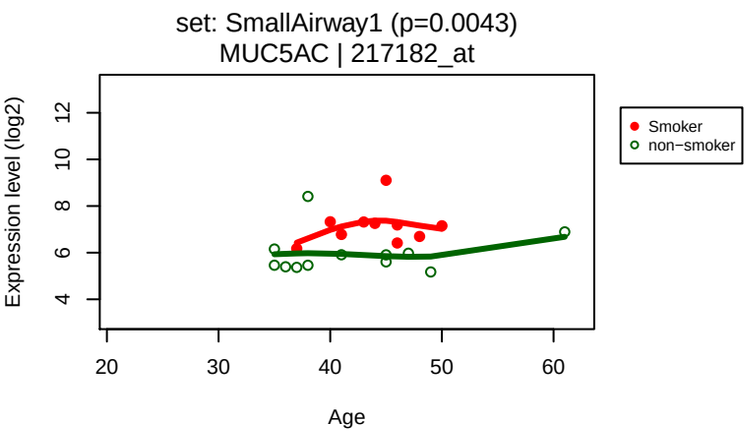
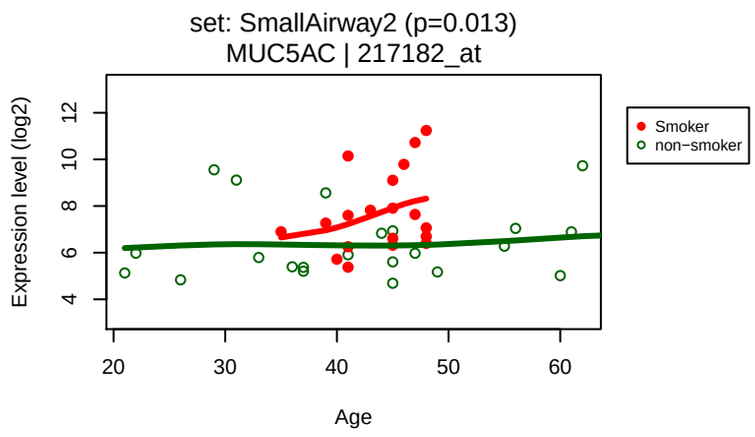
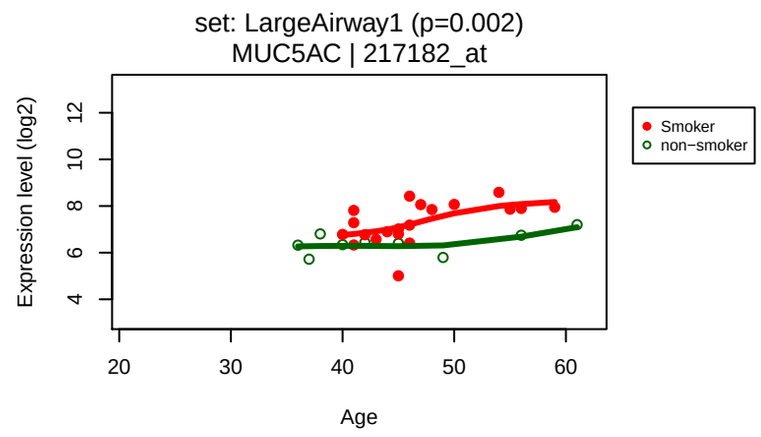

Gene expression of smokers and nonsmokers
gene: MUC5AC | 217182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0043
p=0.013
p=0
p=0.002
p=0.2525
p=0.0048
p=0.613
p=0.267
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

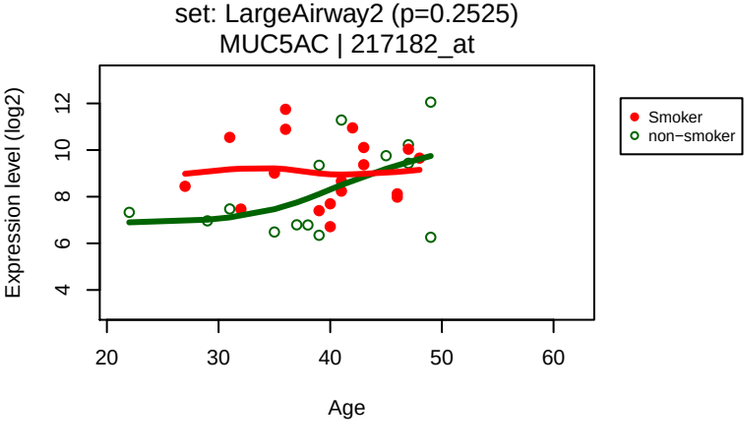

Gene expression ratio (smokers/nonsmokers)
gene: MUC5AC | 217182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0043
p=0.013
p=0
p=0.002
p=0.2525
p=0.0048
p=0.613
p=0.267
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

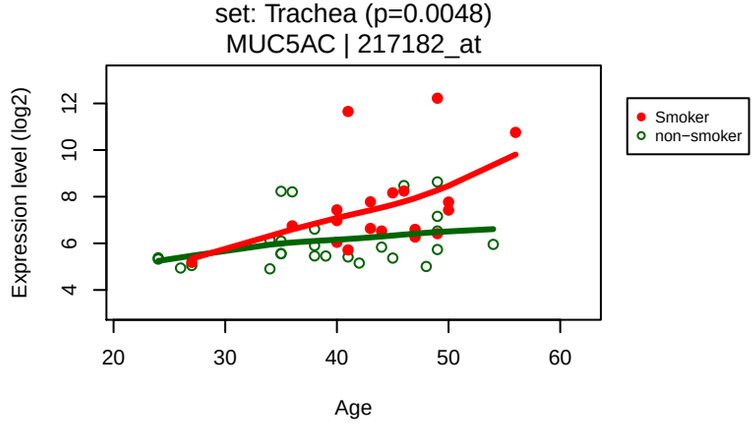

Gene expression ratio in Males
gene: MUC5AC | 217182_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0325
p=0.0827
p=0.0209
p=0.5744
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC5AC | 217182_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1734
p=0.5089
p=0.1324
p=0.3621
SmallAirway2
LargeAirway2
Trachea
Macrophage2

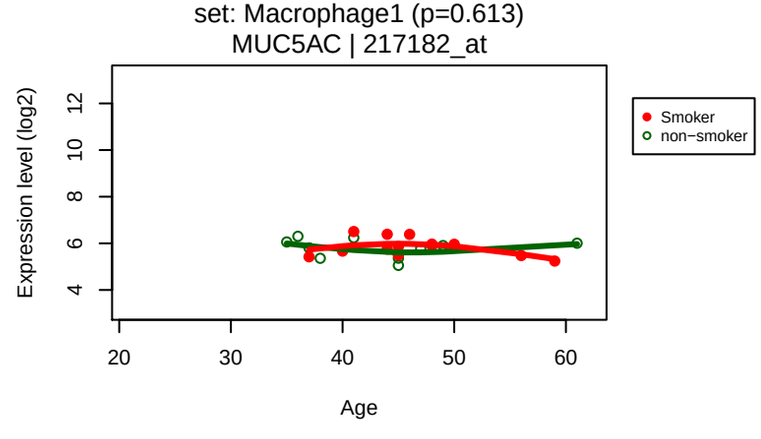

set: SmallAirway1 (p=0.0043)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker

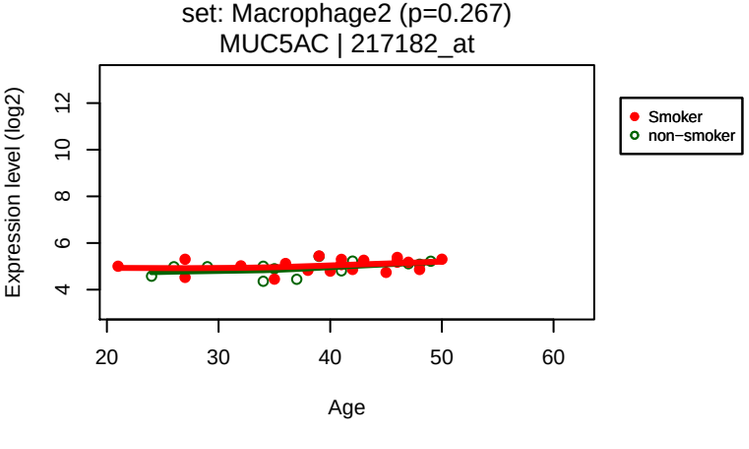

set: SmallAirway2 (p=0.013)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker

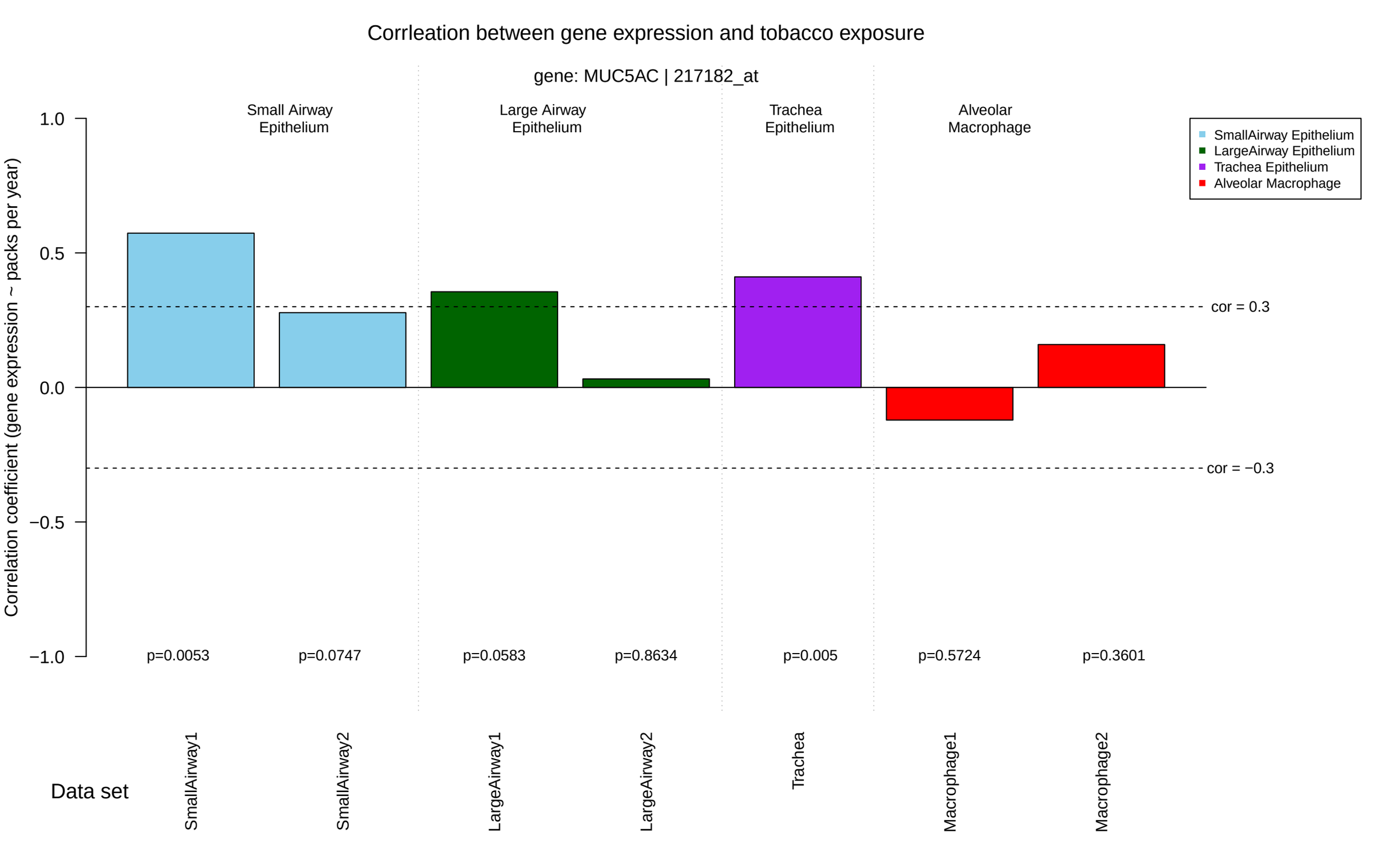

set: LargeAirway1 (p=0.002)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2525)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0048)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.613)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.267)
MUC5AC | 217182_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC5AC | 217182_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0053
p=0.0747
p=0.0583
p=0.8634
p=0.005
p=0.5724
p=0.3601
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2